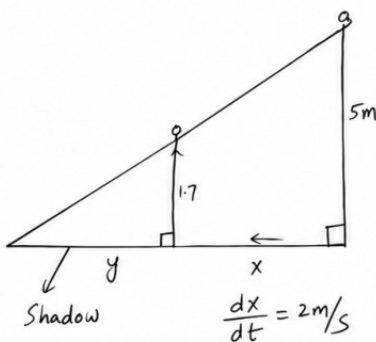


Related Rates continued

60. A street light is on top of a pole which is 5 metres high. A person who is 1.7 metres tall walks away from the pole at a constant rate of 2 metres per second. As he walks his shadow lengthens.

- (a) Find the rate at which the length of the person's shadow is increasing.
 (b) Find the rate at which the tip of the person's shadow is moving away from the pole.



By Similar triangles

(a)

$$\frac{5}{x+y} = \frac{1.7}{y}$$

$$5y = 1.7x + 1.7y$$

$$3.3y = 1.7x$$

$$y = \frac{1.7}{3.3}x$$

$$\frac{d}{dt}(y) = \frac{d}{dt}\left(\frac{1.7}{3.3}x\right)$$

$$\frac{dy}{dt} = \frac{1.7}{3.3} \frac{dx}{dt}$$

$$= \frac{1.7}{3.3} \cdot 2$$

$$\frac{dy}{dt} = \frac{3.4}{3.3} \text{ m/s}$$

rate of person's shadow

(b)

tip of person's shadow:

$$s = x + y$$

$$\frac{d}{dt}(s) = \frac{d}{dt}(x+y)$$

$$\frac{ds}{dt} = \frac{dx}{dt} + \frac{dy}{dt}$$

$$= 2 + \frac{3.4}{3.3}$$

$$= \frac{100}{33} \text{ m/s.}$$

Introduction to Inverse Trigonometric Functions

61. Without using a calculator write down exact values of :

$$\begin{array}{lll}
 \text{(g)} & \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) & \text{(h)} & \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) & \text{(i)} & \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) \\
 \text{(j)} & \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) & \text{(k)} & \tan^{-1}(\sqrt{3}) & \text{(l)} & \tan^{-1}(-\sqrt{3})
 \end{array}$$

$$\begin{array}{ll}
 \text{62} & \begin{array}{l} \sin^{-1}x \\ \cos^{-1}x \end{array} \left\{ \begin{array}{l} -1 \leq x \leq 1 \\ x \in \mathbb{R} \end{array} \right. \\
 \text{(a)} & \sin^{-1}(2) \rightarrow \text{DNE} \\
 \text{(b)} & \cos^{-1}(2) \rightarrow \text{DNE} \\
 \text{(c)} & \tan^{-1}(2) \rightarrow \text{exist} \\
 \text{(d)} & \sin^{-1}(-2) \rightarrow \text{DNE} \\
 \text{(e)} & \tan^{-1}(-2) \rightarrow \text{exist} \\
 \text{(f)} & \sin^{-1}\left(\tan\frac{\pi}{3}\right) = \sin^{-1}(\sqrt{3}) = \text{DNE} \\
 \text{(g)} & \sin^{-1}\left(-\tan\frac{\pi}{6}\right) = \sin^{-1}\left(-\frac{1}{\sqrt{3}}\right) \Rightarrow \text{exist}
 \end{array}$$

$$\begin{array}{ll}
 \text{61} & \\
 \text{(g)} & \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4} \\
 \text{(h)} & \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) = -\frac{\pi}{4} \\
 \text{(i)} & \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6} \\
 \text{(j)} & \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{6} \\
 \text{(k)} & \tan^{-1}(\sqrt{3}) = \frac{\pi}{3} \\
 \text{(l)} & \tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}
 \end{array}$$

62. Without using the calculator state which of the following **do not exist**:

$$\begin{array}{llll}
 \text{(a)} & \sin^{-1} 2 & \text{(b)} & \cos^{-1} 2 \\
 \text{(c)} & \tan^{-1} 2 & \text{(d)} & \sin^{-1}(-2) \\
 \text{(e)} & \tan^{-1}(-2) & \text{(f)} & \sin^{-1}\left(\tan\frac{\pi}{3}\right) \\
 \text{(g)} & \sin^{-1}\left(-\tan\frac{\pi}{6}\right)
 \end{array}$$

$$\begin{array}{ll}
 \text{62} & \begin{array}{l} \sin^{-1}x \\ \cos^{-1}x \end{array} \left\{ \begin{array}{l} -1 \leq x \leq 1 \\ x \in \mathbb{R} \end{array} \right. \\
 \text{(a)} & \sin^{-1}(2) \rightarrow \text{DNE} \\
 \text{(b)} & \cos^{-1}(2) \rightarrow \text{DNE} \\
 \text{(c)} & \tan^{-1}(2) \rightarrow \text{exist} \\
 \text{(d)} & \sin^{-1}(-2) \rightarrow \text{DNE} \\
 \text{(e)} & \tan^{-1}(-2) \rightarrow \text{exist} \\
 \text{(f)} & \sin^{-1}\left(\tan\frac{\pi}{3}\right) = \sin^{-1}(\sqrt{3}) = \text{DNE} \\
 \text{(g)} & \sin^{-1}\left(-\tan\frac{\pi}{6}\right) = \sin^{-1}\left(-\frac{1}{\sqrt{3}}\right) \Rightarrow \text{exist}
 \end{array}$$

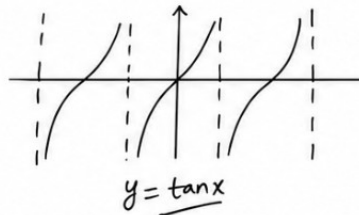
$$\begin{array}{ll}
 \text{61} & \\
 \text{(g)} & \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4} \\
 \text{(h)} & \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) = -\frac{\pi}{4} \\
 \text{(i)} & \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6} \\
 \text{(j)} & \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{6} \\
 \text{(k)} & \tan^{-1}(\sqrt{3}) = \frac{\pi}{3} \\
 \text{(l)} & \tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}
 \end{array}$$

Properties of Inverse Trigonometric Functions

63. Consider the function $y = \tan^{-1}(\tan x)$.

- State the domain.
- State the period.
- Show that the function is odd.
- State the values of x for which $\tan^{-1}(\tan x) = x$
- Hence sketch the graph of the function $y = \tan^{-1}(\tan x)$ for $-\frac{3\pi}{2} \leq x \leq \frac{3\pi}{2}$.

3 $y = \tan^{-1}(\tan x)$



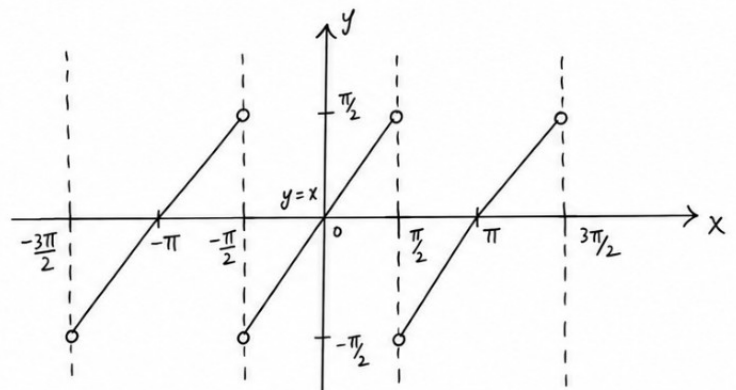
(a) Domain: $x \neq n\pi + \frac{\pi}{2}, n \in \mathbb{Z}$

(b) $f(x) = \tan x$
 $f(x+\pi) = \tan(x+\pi) = \tan x$
 Period is π ✓

(c) $f(x) = \tan^{-1}(\tan x)$
 $f(-x) = \tan^{-1}(\tan(-x)) = \tan^{-1}(-\tan x)$
 $= -\tan^{-1}(\tan x)$
 $= -f(x)$
 $\therefore f(-x) = -f(x)$
 $\therefore f(x)$ is an odd function.

(d) $\tan^{-1}(\tan x) = x$
 when $-\frac{\pi}{2} < x < \frac{\pi}{2}$

(e) $y = \tan^{-1}(\tan x)$ for $-\frac{3\pi}{2} \leq x \leq \frac{3\pi}{2}$



Properties of Inverse Trigonometric Functions continued

64. (a) Prove that $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$.
- (b) Prove that $\cos^{-1}\left(\frac{3}{\sqrt{10}}\right) + \cos^{-1}\left(\frac{2}{\sqrt{5}}\right) = \frac{\pi}{4}$.
- (c) Prove that $\tan^{-1}\frac{3}{4} = 2 \tan^{-1}\frac{1}{3}$.

64 (a) Prove that $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$

$$\alpha = \tan^{-1}\left(\frac{1}{4}\right) \quad \beta = \tan^{-1}\left(\frac{3}{5}\right)$$

$$\Leftrightarrow \begin{cases} \tan \alpha = \frac{1}{4} > 0 \\ 0 < \alpha < \frac{\pi}{2} \end{cases} \quad \Leftrightarrow \begin{cases} \tan \beta = \frac{3}{5} > 0 \\ 0 < \beta < \frac{\pi}{2} \end{cases}$$

$$\alpha + \beta = \frac{\pi}{4}$$

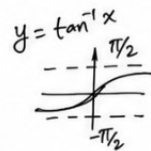
$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta} = \frac{\frac{1}{4} + \frac{3}{5}}{1 - \frac{1}{4} \cdot \frac{3}{5}} = \frac{\frac{5+12}{20}}{\frac{20-3}{20}} = \frac{17}{17} = 1$$

$$\tan(\alpha + \beta) = 1$$

$$\tan(\alpha + \beta) = \tan \frac{\pi}{4}$$

$$\therefore \alpha + \beta = \frac{\pi}{4}$$

hence proved



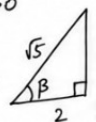
(b) Prove that $\cos^{-1}\left(\frac{3}{\sqrt{10}}\right) + \cos^{-1}\left(\frac{2}{\sqrt{5}}\right) = \frac{\pi}{4}$

$$\alpha = \cos^{-1}\left(\frac{3}{\sqrt{10}}\right) \quad \beta = \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$$

$$\Leftrightarrow \begin{cases} \cos \alpha = \frac{3}{\sqrt{10}} > 0 \\ 0 < \alpha < \frac{\pi}{2} \end{cases} \quad \Leftrightarrow \begin{cases} \cos \beta = \frac{2}{\sqrt{5}} > 0 \\ 0 < \beta < \frac{\pi}{2} \end{cases}$$



$$\sin \alpha = \frac{1}{\sqrt{10}}$$



$$\sin \beta = \frac{1}{\sqrt{5}}$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$= \frac{3}{\sqrt{10}} \cdot \frac{2}{\sqrt{5}} - \frac{1}{\sqrt{10}} \cdot \frac{1}{\sqrt{5}}$$

$$= \frac{6-1}{\sqrt{50}} = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\cos(\alpha + \beta) = \frac{1}{\sqrt{2}}$$

$$\cos(\alpha + \beta) = \cos \frac{\pi}{4} \Rightarrow \alpha + \beta = \frac{\pi}{4} \quad \text{hence proved}$$

(c) Prove that $\tan^{-1}\frac{3}{4} = 2 \tan^{-1}\frac{1}{3}$

$$\alpha = \tan^{-1}\left(\frac{1}{3}\right)$$

$$\Leftrightarrow \begin{cases} \tan \alpha = \frac{1}{3} \\ 0 < \alpha < \frac{\pi}{2} \end{cases} \quad \text{Prove: } \tan^{-1}\frac{3}{4} = 2\alpha$$

$$\Leftrightarrow \tan 2\alpha = \frac{3}{4}$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$= \frac{2(\frac{1}{3})}{1 - (\frac{1}{3})^2} = \frac{\frac{2}{3}}{\frac{8}{9}}$$

$$= \frac{2}{3} \times \frac{9}{8} = \frac{3}{4}$$

$$\tan 2\alpha = \frac{3}{4}$$

$$\Rightarrow 2\alpha = \tan^{-1}\left(\frac{3}{4}\right)$$

Differentiation of Inverse Trigonometric Functions

65. Assuming all the functions below are differentiable, find the derivatives of :

(a) $\sin^{-1}(x^2)$

(b) $\cos^{-1}\left(\frac{x}{2}\right)$

(c) $\tan^{-1}\left(\frac{3-x}{5}\right)$

(d) $e^x \sin^{-1}(x)$

(e) $\ln[\tan^{-1}(x)]$

(f) $\tan^{-1}\left(\frac{x+2}{1-2x}\right)$

65
(a) $(\sin^{-1}(x^2))' = \frac{2x}{\sqrt{1-x^4}}$

(d) $(e^x \cdot \sin^{-1}x)'$
 $= e^x \cdot \frac{1}{\sqrt{1-x^2}} + \sin^{-1}x \cdot e^x$
 $= e^x \left(\frac{1}{\sqrt{1-x^2}} + \sin^{-1}x \right)$

(f) $(\tan^{-1}(\frac{x+2}{1-2x}))' = \frac{(1-2x) \cdot 1 - (x+2)(-2)}{(1-2x)^2}$
 $= \frac{1-2x+2x+4}{(1-2x)^2 + (x+2)^2}$
 $= \frac{5}{1-4x+4x^2+x^2+4x+4}$
 $= \frac{5}{5x^2+5}$
 $= \frac{1}{x^2+1}$

(b) $(\cos^{-1}(\frac{x}{2}))' = \frac{-1}{\sqrt{4-x^2}}$

(c) $(\tan^{-1}(\frac{3-x}{5}))' = \frac{-\frac{1}{5}}{1+(\frac{3-x}{5})^2}$
 $= \frac{-\frac{1}{5}}{\frac{25+(3-x)^2}{25}} = \frac{-5}{25+(3-x)^2}$

(e) $(\ln(\tan^{-1}x))'$
 $= \frac{\frac{1}{1+x^2}}{\tan^{-1}x} = \frac{1}{(1+x^2)\tan^{-1}x}$

Differentiation of Inverse Trigonometric Functions continued

66. Find $f''(x)$ for each of the following functions:

(a) $f(x) = \sin^{-1} x$

(b) $f(x) = \sin^{-1} \frac{x}{4}$

(c) $f(x) = \tan^{-1}(x^3)$

① $f(x) = \sin^{-1} x$

$$f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$f'(x) = (1-x^2)^{-\frac{1}{2}}$$

$$f''(x) = -\frac{1}{2} (1-x^2)^{-\frac{3}{2}} \cdot (-2x)$$

$$f''(x) = \frac{x}{(1-x^2)^{\frac{3}{2}}}$$

② $f(x) = \sin^{-1}\left(\frac{x}{4}\right)$

$$f'(x) = \frac{1}{\sqrt{16-x^2}}$$

$$f'(x) = (16-x^2)^{-\frac{1}{2}}$$

$$f''(x) = -\frac{1}{2} (16-x^2)^{-\frac{3}{2}} \cdot (-2x)$$

$$f''(x) = \frac{x}{(16-x^2)^{\frac{3}{2}}}$$

③ $f(x) = \tan^{-1}(x^3)$

$$f'(x) = \frac{3x^2}{1+(x^3)^2}$$

$$f'(x) = \frac{3x^2}{1+x^6} \left(\frac{u}{v}\right)$$

$$f''(x) = \frac{(1+x^6) \cdot (6x) - 3x^2(6x^5)}{(1+x^6)^2}$$

$$= \frac{6x(1+x^6-3x^6)}{(1+x^6)^2} = \frac{6x(1-2x^6)}{(1+x^6)^2}$$

67. Find $\frac{dy}{dx}$ for each of the following :

(a) $\cos^{-1} x + \sin^{-1} y = \pi$

(b) $\tan^{-1}(2y) = x^2 + y$

(c) $\tan^{-1}(xy) = y$

67
① $\cos^{-1} x + \sin^{-1} y = \pi$

$$\frac{d}{dx}(\sin^{-1} y) = \frac{d}{dx}(\pi - \cos^{-1} x)$$

$$\frac{1}{\sqrt{1-y^2}} \frac{dy}{dx} = 0 - \left(\frac{-1}{\sqrt{1-x^2}}\right)$$

$$\frac{1}{\sqrt{1-y^2}} \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

② $\frac{d}{dx}(\tan^{-1}(2y)) = \frac{d}{dx}(x^2 + y)$

$$\frac{2 \frac{dy}{dx}}{1+(2y)^2} = 2x + \frac{dy}{dx}$$

$$\left(\frac{2}{1+4y^2} - 1\right) \frac{dy}{dx} = 2x$$

$$\left(\frac{2-1-4y^2}{1+4y^2}\right) \frac{dy}{dx} = 2x$$

$$\left(\frac{1-4y^2}{1+4y^2}\right) \frac{dy}{dx} = 2x$$

$$\frac{dy}{dx} = \frac{2x(1+4y^2)}{1-4y^2}$$

③ $\frac{d}{dx}(\tan^{-1}(xy)) = \frac{d}{dx}(y)$

$$\frac{x \cdot \frac{dy}{dx} + y \cdot 1}{1+(xy)^2} = \frac{dy}{dx}$$

$$x \cdot \frac{dy}{dx} + y = \frac{dy}{dx} (1+x^2y^2)$$

$$y = \frac{dy}{dx} (1+x^2y^2 - x)$$

$$\frac{dy}{dx} = \frac{y}{1+x^2y^2 - x}$$

Applications of Differentiation of Inverse Trigonometric Functions

68. Find the equations of the tangent and the normal to $y = [\sin^{-1} x]^2$ at the point where

$x = \frac{-1}{\sqrt{2}}$. 68 $y = (\sin^{-1} x)^2$ at the point $x = \frac{-1}{\sqrt{2}}$

$$y' = 2(\sin^{-1} x) \cdot \frac{1}{\sqrt{1-x^2}}$$

$$x = \frac{-1}{\sqrt{2}}, m_T = 2 \left[\sin^{-1} \left(\frac{-1}{\sqrt{2}} \right) \right] \cdot \frac{1}{\sqrt{1 - \left(\frac{-1}{\sqrt{2}} \right)^2}}$$

$$= 2 \left(-\frac{\pi}{4} \right) \cdot \frac{1}{\left(\frac{1}{\sqrt{2}} \right)}$$

$$= -\frac{\pi}{2} \cdot \sqrt{2}$$

$$m_T = -\frac{\pi\sqrt{2}}{2}$$

$$m_N = -\frac{1}{m_T} = \frac{+1}{\frac{\pi\sqrt{2}}{2}} = \frac{2}{\pi\sqrt{2}} = \frac{\sqrt{2}}{\pi}$$

Point: $x = \frac{-1}{\sqrt{2}}$
 $y = \left(\sin^{-1} \left(\frac{-1}{\sqrt{2}} \right) \right)^2 = \left(-\frac{\pi}{4} \right)^2 = \frac{\pi^2}{16}$
 $\left(\frac{-1}{\sqrt{2}}, \frac{\pi^2}{16} \right)$

Equation of the tangent at $\left(\frac{-1}{\sqrt{2}}, \frac{\pi^2}{16} \right)$ is

$$y - \frac{\pi^2}{16} = -\frac{\pi\sqrt{2}}{2} \left(x + \frac{1}{\sqrt{2}} \right)$$

$$y = -\frac{\pi\sqrt{2}}{2} x - \frac{\pi}{2} + \frac{\pi^2}{16}$$

Equation of the Normal at $\left(\frac{-1}{\sqrt{2}}, \frac{\pi^2}{16} \right)$ is

$$y - \frac{\pi^2}{16} = \frac{\sqrt{2}}{\pi} \left(x + \frac{1}{\sqrt{2}} \right)$$

$$y = \frac{\sqrt{2}}{\pi} x + \frac{1}{\pi} + \frac{\pi^2}{16}$$

69. (a) Prove that $\frac{d}{dx} \left[\tan^{-1}(x) + \tan^{-1} \left(\frac{1}{x} \right) \right] = 0$ for $x > 0$, and hence show that

$$\tan^{-1}(x) + \tan^{-1} \left(\frac{1}{x} \right) = \frac{\pi}{2}, \text{ for } x > 0.$$

(b) Find the value of $\tan^{-1}(x) + \tan^{-1} \left(\frac{1}{x} \right)$ for $x < 0$.

69 a $\frac{d}{dx} \left[\tan^{-1}(x) + \tan^{-1} \left(\frac{1}{x} \right) \right] = 0$ for $x > 0$ b Find the Value of $\left[\tan^{-1}(x) + \tan^{-1} \left(\frac{1}{x} \right) \right]$ for $x < 0$

Let $f(x) = \tan^{-1}(x) + \tan^{-1} \left(\frac{1}{x} \right)$, $x > 0$

$$f'(x) = \frac{1}{1+x^2} + \frac{\left(\frac{-1}{x^2} \right)}{1 + \left(\frac{1}{x} \right)^2} \cdot \frac{x^2}{x^2}$$

$$= \frac{1}{1+x^2} - \frac{1}{x^2+1}$$

$$f'(x) = 0, x > 0 \Rightarrow \frac{d}{dx} \left[\tan^{-1}(x) + \tan^{-1} \left(\frac{1}{x} \right) \right] = 0$$

$\therefore f(x)$ is a constant when $x > 0$

$$x=1, f(1) = \tan^{-1}(1) + \tan^{-1} \left(\frac{1}{1} \right)$$

$$= \frac{\pi}{4} + \frac{\pi}{4}$$

$$= \frac{\pi}{2}$$

$$\therefore \tan^{-1}(x) + \tan^{-1} \left(\frac{1}{x} \right) = \frac{\pi}{2} \text{ for } x > 0$$

Let $f(x) = \tan^{-1}(x) + \tan^{-1} \left(\frac{1}{x} \right)$

Let $x = -y$, $y > 0$

$$f(-y) = \tan^{-1}(-y) + \tan^{-1} \left(\frac{1}{-y} \right)$$

$$= -\tan^{-1}(y) - \tan^{-1} \left(\frac{1}{y} \right)$$

$$= - \left[\tan^{-1}(y) + \tan^{-1} \left(\frac{1}{y} \right) \right], y > 0$$

$$f(-y) = -\frac{\pi}{2} \text{ (from (a))}$$

$$\therefore f(x) = -\frac{\pi}{2}$$

$$\therefore \tan^{-1}(x) + \tan^{-1} \left(\frac{1}{x} \right) = -\frac{\pi}{2}, x < 0.$$

Symmetric Properties

1. $\sin^{-1}(-x) = -\sin^{-1} x$
2. $\cos^{-1}(-x) = \pi - \cos^{-1} x$
3. $\tan^{-1}(-x) = -\tan^{-1} x$

Applications of Differentiation of Inverse Trigonometric Functions continued

70. (a) Show that the curve $y = x \sin^{-1} x$ has a stationary point at $(0,0)$ and determine its nature. You may assume that this is the only stationary point.
- (b) Show that the function $y = x \sin^{-1} x$ is even.
- (c) Find the domain and range of the function $y = x \sin^{-1} x$.
- (d) Hence sketch the graph of the function $y = x \sin^{-1} x$.

70

(a)

$$y = \frac{x \sin^{-1} x}{1}$$

$$y'(x) = x \cdot \frac{1}{\sqrt{1-x^2}} + \sin^{-1} x \cdot 1$$

$$y'(x) = \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x$$

Substitute $(0,0)$

$$y'(0) = \frac{0}{\sqrt{1-0^2}} + \sin^{-1} 0$$

$$y'(0) = 0 + 0$$

$$y'(0) = 0$$

$\therefore (0,0)$ is the stationary point.

Nature:

$$y'(x) = \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x$$

x	$-\frac{1}{2}$	0	$\frac{1}{2}$
$y'(x)$	-	0	+

Minimum at $(0,0)$.

(b) $f(x) = x \sin^{-1} x$

$$\begin{aligned} f(-x) &= (-x) \sin^{-1}(-x) \\ &= -x (-\sin^{-1} x) \\ &= x \sin^{-1} x \end{aligned}$$

$$f(-x) = f(x)$$

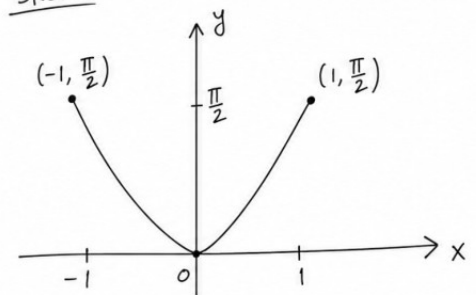
$\therefore f(x)$ is an even function.

(c) $y = x \sin^{-1} x$

Domain: $[-1, 1]$ ✓

Range: $x = -1, y = (-1) \sin^{-1}(-1) = \frac{\pi}{2}$
 $x = 1, y = 1 \cdot \sin^{-1}(1) = \frac{\pi}{2}$
 $x = 0, y = 0 \cdot \sin^{-1}(0) = 0$
 $[0, \frac{\pi}{2}]$ ✓

(d) sketch



Applications of Differentiation of Inverse Trigonometric Functions continued

71. A point P is observed on a vertical pole 10 metres above the level of an observer's eye. Q is a point 30 metres above P . If the observer is x metres from the pole, prove that the angle β subtended by PQ at the observer's eye is given by

$$\beta = \tan^{-1}\left(\frac{40}{x}\right) - \tan^{-1}\left(\frac{10}{x}\right).$$

Hence, or otherwise, find where the observer must stand in order for this angle β to be greatest.

$\beta = \alpha - \gamma$
 $\tan \alpha = \frac{AQ}{OA} = \frac{40}{x}$
 $0 < \alpha < \frac{\pi}{2}$

$\tan \gamma = \frac{AP}{OA} = \frac{10}{x}$
 $0 < \gamma < \frac{\pi}{2}$

$\therefore \beta = \tan^{-1}\left(\frac{40}{x}\right) - \tan^{-1}\left(\frac{10}{x}\right) \checkmark$

$\beta(x) = \tan^{-1}\left(\frac{40}{x}\right) - \tan^{-1}\left(\frac{10}{x}\right)$

$\beta'(x) = \frac{\left(\frac{-40}{x^2}\right)}{1 + \left(\frac{40}{x}\right)^2} - \frac{\left(\frac{-10}{x^2}\right)}{1 + \left(\frac{10}{x}\right)^2}$

$\beta'(x) = \frac{-40}{1600 + x^2} + \frac{10}{100 + x^2}$

Stationary Points:

$\beta'(x) = 0$

$\frac{-40}{1600 + x^2} + \frac{10}{100 + x^2} = 0$

$\frac{40}{1600 + x^2} = \frac{10}{100 + x^2}$

$4(100 + x^2) = 1600 + x^2$

$400 + 4x^2 = 1600 + x^2$

$3x^2 = 1200$

$x^2 = 400$

$x = \pm 20$ [$\because x > 0$]

$x = 20\text{m} \checkmark$

Nature: $\beta'(x) = \frac{-40}{1600 + x^2} + \frac{10}{100 + x^2}$

x	10	20	30
$\beta'(x)$	+	0	-

$\beta(x)$ is maximum when $x = 20\text{m}$.

Maximum angle of β is

$x = 20$

$\beta(20) = \tan^{-1}\left(\frac{40}{20}\right) - \tan^{-1}\left(\frac{10}{20}\right)$

$\beta = \tan^{-1}(2) - \tan^{-1}\left(\frac{1}{2}\right)$

$\tan \beta = \tan\left[\tan^{-1}(2) - \tan^{-1}\left(\frac{1}{2}\right)\right]$

$= \frac{2 - \frac{1}{2}}{1 + 2 \cdot \frac{1}{2}}$

$= \frac{\frac{3}{2}}{2}$

$\tan \beta = \frac{3}{4}$

$0 < \beta < \frac{\pi}{2}$

$\beta = \tan^{-1}\left(\frac{3}{4}\right)$

greatest angle of β

Applications of Differentiation of Inverse Trigonometric Functions continued

72. A plane maintains a constant altitude of 6000 metres whilst flying horizontally away from an airport at a constant speed of 600 km/hr. Let x km be the horizontal distance of the plane from the airport and let θ be the angle of elevation of the plane from the airport.

- (a) Show that $\theta = \tan^{-1} \frac{6}{x}$.
- (b) Hence find, in radians per second, the rate at which θ is decreasing when $x = 3$.

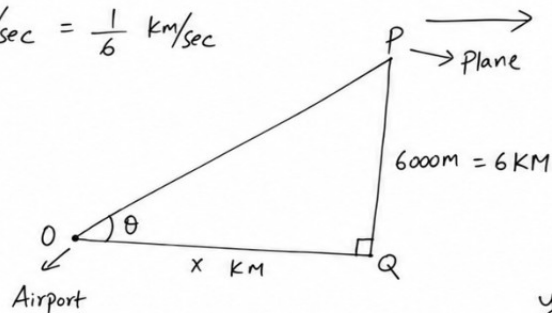
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Given $\frac{dx}{dt} = 600 \text{ km/hr} = \frac{600}{3600} \text{ km/sec} = \frac{1}{6} \text{ km/sec}$

(a)

$$\tan \theta = \frac{PQ}{OQ}$$

$$\left. \begin{array}{l} \tan \theta = \frac{6}{x} \\ 0 < \theta < \frac{\pi}{2} \end{array} \right\} \Rightarrow \theta = \tan^{-1} \frac{6}{x}$$



(b)

find $\frac{d\theta}{dt}$ when $x = 3 \text{ km}$

$$\begin{aligned} \theta &= \tan^{-1} \frac{6}{x} \\ \frac{d}{dt}(\theta) &= \frac{d}{dt} \left(\tan^{-1} \frac{6}{x} \right) \\ \frac{d\theta}{dt} &= \frac{-\frac{6}{x^2}}{1 + \left(\frac{6}{x}\right)^2} \cdot \frac{dx}{dt} \\ &= \frac{-6}{x^2 + 36} \cdot \frac{1}{6} \end{aligned}$$

$$\frac{d\theta}{dt} = \frac{-1}{x^2 + 36}$$

$$x = 3, \quad \frac{d\theta}{dt} = \frac{-1}{3^2 + 36} = \frac{-1}{45} \text{ rad/sec}$$

$\therefore \theta$ is decreasing at a rate of $\frac{1}{45} \text{ rad/sec}$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + (f(x))^2}$$

$$y = \tan^{-1} x$$

$$-\frac{\pi}{2} < y < \frac{\pi}{2}$$